INCH-POUND
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DETAIL SPECIFICATION

LEATHER, CATTLEHIDE, UPHOLSTERY

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for cattlehide leather used on upholstery and motor vehicle seats.

1.2 Classification. The leather is available in the following types and finishes as specified (6.2):

1.2.1 Types.

Type I - Full grain
Type II - Hand buff
Type III - Machine buff
Type IV - Deep buff

1.2.2 Finishes.

Finish A - Smooth grain
Finish B - Boarded (bend or machine)
Finish C - Embossed in fine grain
Finish D - Embossed in coarse grain
Finish E - Equal to an approved sample

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AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1813 - Standard Test Method for Measuring Thickness of Leather Specimens
ASTM D1814 - Standard Test Method for Measuring Thickness of Leather Units
ASTM D2208 - Standard Test Method for Breaking Strength of Leather by Grab Method
ASTM D2810 - Standard Test Method for pH of Leather
ASTM D2813 - Standard Practice for Sampling Leather for Physical and Chemical Tests
ASTM D3495 - Standard Test Method for Hexane Extraction of Leather

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ASTM D5053 - Standard Test Method for Resistance to Crocking of Leather
ASTM D6075 - Standard Test Method for Cracking Resistance of Leather

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. The material shall be cattlehide leather. The flesh side shall be smooth and free of coarse loose fibers and flesh.

3.3.1 Type I, full grain. The leather shall be a hide from which nothing except the hair and scud has been removed from the grain side. Superficial scars, scratches, and other grain defacements may be lightly buffed by hand, but not by machine. The total area so affected shall be not more than 5 percent of the total area of the hide.

3.3.2 Type II, hand buff. The leather shall be a full grain hide except that the entire grain area may be scuffed, but not more than the depth required to remove only the top portions of the hair follicles.

3.3.3 Type III, machine buff. The leather shall be a hide from which not more than the grain layer has been removed, leaving no characteristics of the grain pattern on any part of the underlying surface.

3.3.4 Type IV, deep buff. The leather shall new hide from which not more than the grain layer has been removed, leaving no characteristics of the grain pattern on any part of the underlying surface.

3.4 Tannage. The leather shall be tanned by a vegetable, mineral or synthetic tanning process, or a suitable combination of such processes.

3.5 Color. The color of the finished leather shall be as specified (see 6.2). For Types I and II leather, the color (dye) shall penetrate not less than the full depth of the grain leather. For Types III and IV leather, the color (dye) shall penetrate the grain side of the leather to a depth

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of not less than two (2) ounces. (Note: 1 ounce =1/64 inch thick). Visual examination of the color and appearance shall be tested as specified in 4.6.1.

3.6 Physical requirements.

3.6.1 Trim (form and area). The leather shall be whole hides, evenly trimmed, with the head cut off behind the natural openings, the shanks cut off above the knees, and the tail not more than two (2) inches long. Unless otherwise specified (see 6.2 and 6.4) the minimum area of each hide shall be 36 square feet. The maximum area of each hide shall be 55 square feet. The area, expressed in terms of square feet, shall be legible marked in the butt area of the flesh side of each hide leather.

3.6.2 Finish. The grain surface of the leather shall be smooth, boarded or embossed as required for the finish specified (see 6.2). The finish coat shall be a suitable material that is securely anchored to the grains of the surface of the leather. When finish "E" is specified, the finish of the grain surface shall be equal to an approved sample (see 6.3).

3.6.3 Thickness. Unless otherwise specified (see 6.2), the leather shall be 3 (± 1/2) ounces in thickness. The flesh side may be buffed shaved or skived to produce a uniform thickness.

3.6.4 Cracking, low temperature. The leather shall not crack when tested as specified in 4.5.

3.6.5 Colorfastness to crocking. The finished leather shall not show a rating less than 4.0 for dry crocking and not less than 3.0 for wet crocking when tested as specified in 4.5.

3.6.6 Tackiness. The leather shall show no tendency toward tackiness when tested as specified in 4.5.

3.6.7 Breaking force and elongation. Not less than 80 percent of the specimens tested shall meet the minimum breaking force values in Table I when tested as specified in 4.5. Not less than 80 of the specimens tested shall have elongation values of maximum of 40 percent at the loads shown in Table I when tested as specified in 4.5.

TABLE I. Breaking force and elongation requirements.

Types	Breaking force, pounds (min.) (ounces) 1/	Elongation load (pounds) 1/
I and II	90	50
III and IV	135	100

1/ Not applicable to leather less than 2 1/2 ounces in thickness.

3.7 Chemical requirements. The leather shall conform to the chemical requirements of Table II when tested as specified in 4.5.

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TABLE II. Chemical requirements.

Characteristic	Requirement
Hexane soluble material, percent (max.) 1/ Minimum Maximum	 8.0 20.0
Acidity (pH)	3.3 – 5.0

1/ Calculated on a moisture-free basis.

3.8 Workmanship. The finished leather shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5 and 4.7.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination of the end item. The leather for each type shall be examined and defects shall be classified as stated in Table III. The sample unit for these examinations shall be one (1) side, back, bend or double shoulder. The lot size shall be expressed in terms of one (1) side, back, bend or double shoulder. The contract or purchase order shall identify the number of samples required for thickness for acceptance and rejection of the leather skins. Separate examinations shall be performed for visual and the thickness, but the same sample may be used for each examination.

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TABLE III. Visual examination defects.

Examination	Defect
Material	Not cattlehide
Type	Not as specified
Trim	Not whole hide with head removed Not evenly trimmed Shanks not cut off above the knees
Quality of leather	Stain or foreign matter
	Flesh side not smooth or has areas of loose flesh
	Any cut, scratch, hole, band, broken grain, vein, hip mark, wrinkle, rough shoulder, or other imperfections which individually or collectively cause more than six (6) percent loss of the total area of hide.
Color (all types)	Not as specified Dye does not completely penetrate leather Dye penetrates grain side to a depth of less than two (2) ounces
Area (expressed in terms of square feet)	Not as marked on hide Unless otherwise specified (see 3.6.1), area less than 36 square feet Area greater than 55 square ft.
Marking	Omitted, eligible, or not in area specified
Thickness 1/	Not as specified

1/ Thickness shall be measured in accordance with ASTM D1814.

4.5 End item testing. The finished leather shall be tested for the characteristics as specified in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. Sample grinding and sampling location from which the sample unit is to be obtained shall be in accordance with ASTM D2813. The test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exists:

a. More than three (3) specimen failures occur for breaking force and elongation.
b. More than one (1) specimen fails to meet any of the remaining requirements applicable to the sample unit.
c. Any composite fails to meet the specified requirement.

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TABLE IV. End item tests.

Characteristic	Requirement Reference	Test Method	Requirement	
			Sample unit	Composite sample
Material identification	See 3.3	Visual (see 6.4)		-
Tannage	See 3.4	Visual (see 6.4)		-
Color	See 3.5	4.6.1		-
Thickness	3.6.3	ASTM D1813	X	-
Cracking	3.6.4	ASTM D6075 and 4.6.2	X	-
Colorfastness to crocking 1/	3.6.5	ASTM D5053	X	-
Tackiness	3.6.6	ASTM D6183	X	-
Breaking force	3.6.7 & Table I	ASTM D2208	X	-
Elongation	3.6.7 & Table I	ASTM D2208	X	-
Hexane-soluble material	Table II	ASTM D3495	-	X
Acidity (pH)	Table II	ASTM D2810	-	X

1/ Evaluated using AATCC EP8.

4.6 Methods of testing and inspection.

4.6.1 Visual shade examination. The finished leather shall be viewed using AATCC Evaluation Procedure 9, Option C when viewed under filtered tungsten lamps which approximate artificial daylight D_{75} illuminant having a correlated color temperature of 7500K (± 200 K) with illumination of 100 (± 20) foot- candles, and shall be a good match to the standard sample under incandescent lamplight at 2850 K (± 200 K).

4.6.2 Cracking. The leather shall be tested for cracking in accordance with ASTM D6075, except that the specimen shall be four (4) by eight (8) inch rectangle of leather and shall be folded twice upon itself in accordance with the following procedure: the specimen shall be folded upon itself along the length dimension with the flesh side in and then folded at right angles to the original fold.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The strap leather is intended primarily for use in post office mail bags, cases, belts, holsters, shoulder straps, and cap visors.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and finish required (see 1.2)
c. The specific issue of individual documents referenced
d. When first article is required (see 3.1)
e. Color required (see 3.5)
f. Conformance inspection (see 4.3)
g. Inspection conditions (see 4.3.1)
h. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Subject term (key word) listing.

Deep buff
Embossed
Full grain
Hand buff
Machine buff

6.6 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army – GL
Air Force – 11

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DLA-CT

(Project: 8330-2022-004)

Review activities:
Army – AT, MD
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